ANSWER KEY

Mid-term test • CSC354

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1. In supervised learning for classification problems in machine learning, which algorithm is commonly ...

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is used for binary classification problems and can be extended to multiclass by using techniques like one-vs-rest (OvR) or softmax.

2. Which supervised learning technique is most suitable for solving a binary classification problem?

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is designed for binary classification problems where the output is a probability that can be mapped to two or more discrete classes.

3. Which of the following is a common supervised learning technique used for solving classification pro...

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is a statistical method for predicting binary outcomes from independent variables and has been widely used in classification problems.

4. In supervised learning for classification problems in machine learning, which algorithm is commonly ...

   ' **Answer: B. Logistic Regression**

   Explanation: In supervised learning for classification problems, Logistic Regression is commonly used to predict the class of given input data by estimating probabilities.

5. Which supervised learning technique is most commonly used for solving binary classification problems...

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is a statistical method for predicting binary outcomes based on one or more independent variables. It estimates the probability that an instance belongs to a particular class.

6. In supervised learning for classification problems, which algorithm is commonly used to predict cate...

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is used for binary classification problems where the outcome is categorical.

7. In supervised learning for classification problems in machine learning, which algorithm is commonly ...

   ' **Answer: B. Logistic Regression**

   Explanation: Both Logistic Regression (when the dependent variable is binary) and Decision Trees are commonly used for classification problems in supervised learning. They predict categorical outcomes based on input features.

8. In supervised learning for classification problems in machine learning, which algorithm is commonly ...

   ' **Answer: B. Logistic Regression**

   Explanation: Logistic regression is used for binary classification problems where the output can be one of two classes.

9 Which supervised learning technique is most suitable for solving a binary classification problem wit...

' **Answer: B. Logistic Regression**

Explanation: Logistic regression is a suitable technique for binary classification problems. It can be adjusted with class weights to handle imbalanced classes effectively.

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10 Which supervised learning technique is most suitable for solving a binary classification problem whe...

' **Answer: B. Logistic Regression with class weights adjustment**

Explanation: When dealing with imbalanced classes in a binary classification problem, Logistic Regression can be adapted by assigning different class weights to handle the imbalance effectively.
